Comparación de los resultados de CIBERSORTx y Bayesprism

1. Cargar paquetes necesarios

```
library(tidyverse)
library(dbplyr)
library(SummarizedExperiment)
library(ggpubr)
library(rstatix)
library(ggpubr)
library(patchwork)
```

2. Cargar y preparar los archivos

```
# -------- BAYESPRISM ---------

bayesprism <- readRDS("BP_resultados2.RDS")
ivygap <- readRDS ("IVYGAP.RDA")
samples_df <- colData(ivygap) %>%
  as.data.frame() %>%
  dplyr::select(Sample, Tumor, Structure, SubtypeW)

#Convertir los datos a formato largo

bayesprism <- bayesprism %>%
  rownames_to_column(var = "muestra")

bayesprism_lf <- bayesprism %>%
  pivot_longer(cols = -muestra, names_to = "tipo_celular", values_to = "proporcion")

#Hay que unir los metadatos de ivygap y los resultados de la deconvolución a
#través de las columnas muestra y samples (columnas en común)

#Hay que asegurarse de que tengan el mismo formato antes de hacer el join:
samples_df$Sample <- as.character(samples_df$Sample)
bayesprism_lf$muestra <- as.character(bayesprism_lf$muestra)

bayesprism_lf_w_structure <- bayesprism_lf %>%
  left_join(samples_df,
    by = c("muestra" = "Sample"))

# ------ CIBERSORTX -------

#Quitar P-Value, Correlation y RMSE para que no se grafiquen como tipos celulares
resultado_cibersort <- read_tsv("deconvolución_IVY_CPM_cibersort.txt", col_names = T)
resultado_cibersort <- resultado_cibersort[, !names(resultado_cibersort)
                                           %in% c("P-value", "Correlation", "RMSE")]
```

```r
#Convertir los datos al formato largo
cibersort_lf <- resultado_cibersort %>%
  pivot_longer(cols = -1, names_to = "tipo_celular", values_to = "proporcion")

#Unir a los metadatos de ivygap
colData(ivygap) -> samples
colnames(samples)

cibersort_lf %>% mutate(Mixture = factor(Mixture)) %>%
  left_join(samples %>% as.data.frame %>%
              dplyr::select(Sample, Tumor, Structure, SubtypeW) %>%
              mutate(Sample = factor(Sample)),
            by = c(Mixture = "Sample")) -> cibersort_lf_w_structure

if ("Mixture" %in% colnames(cibersort_lf_w_structure)) {
  cibersort_lf_w_structure <- cibersort_lf_w_structure %>%
    dplyr::rename(muestra = Mixture)}
```

Tenemos 2 data frames con los que trabajar: cibersort_lf_w_structure y bayesprism_lf_w_structure

3. Combinar data frames

```r
if ("proporcion" %in% colnames(cibersort_lf_w_structure)) {
  cibersort_lf_w_structure <- cibersort_lf_w_structure %>%
    dplyr::rename(proporcion_cibersort = proporcion) }

cibersort_lf_w_structure <- cibersort_lf_w_structure %>%
  mutate(muestra = as.character(muestra),
         Structure = as.factor(Structure))

if ("proporcion" %in% colnames(bayesprism_lf_w_structure)) {
  bayesprism_lf_w_structure <- bayesprism_lf_w_structure %>%
    dplyr::rename(proporcion_bayesprism = proporcion)}

bayesprism_lf_w_structure <- bayesprism_lf_w_structure %>%
  mutate(muestra = as.character(muestra),
         Structure = as.factor(Structure))
```

4. Gráficos de células neoplásicas

```r
# Tipos celulares neoplásicos seleccionados en orden alfabético
tipos_seleccionados <- sort(c("AClike", "OPClike", "MES1like", "MES2like",
                              "NPC1like", "NPC2like"))

# Unir los datos y filtrar por tipo celular
comparacion_con_estructura_pearson <- inner_join(cibersort_lf_w_structure,
                                                  bayesprism_lf_w_structure,
                                                  by = c("muestra", "tipo_celular",
                                                         "Structure", "SubtypeW")) %>%

  filter(tipo_celular %in% tipos_seleccionados) %>%
  mutate(tipo_celular = factor(tipo_celular, levels = tipos_seleccionados)) #ordena tipo_celular
```

```r
# Calcular correlaciones de Pearson por estructura y tipo celular
correlaciones_estructura_pearson <- comparacion_con_estructura_pearson %>%
  group_by(Structure, tipo_celular) %>%
  summarise(cor_pearson = cor(proporcion_cibersort, proporcion_bayesprism,
                              method = "pearson"),
    .groups = "drop") %>%
  mutate(tipo_celular = factor(tipo_celular, levels = tipos_seleccionados))

# ----- Gráfico de dispersión ---------
p <- ggplot(comparacion_con_estructura_pearson,
       aes(x = proporcion_cibersort, y = proporcion_bayesprism, color = Structure)) +
  geom_point(alpha = 0.6, size = 1.5) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(Structure ~ tipo_celular, scales = "free") +
  geom_text(data = correlaciones_estructura_pearson,
            aes(x = Inf, y = -Inf,
                label = paste0("r = ", round(cor_pearson, 2))),
            hjust = 1.1, vjust = -0.5,
            inherit.aes = FALSE,
            size = 3, color = "blue") +
  labs(x = "Proporción CIBERSORTx",
       y = "Proporción BayesPrism",
       title = "Correlación de Pearson entre BayesPrism y CIBERSORTx",
       subtitle = "Células neoplásicas") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5),
        legend.position = "bottom")
ggsave(
  filename = "celulasneoplasicas.png",
  plot = p,
  path = "C:/Users/usuario/Documents/1. Practicas/resultados",
  width = 13, height = 6, dpi = 300,
  bg = "white")

print(p)
```

```r
#------ Heatmap  --------

#Ordenar eje Y
correlaciones_estructura_pearson$Structure <- factor(
  correlaciones_estructura_pearson$Structure,
  levels = c("CT", "CTmvp", "CTpan", "IT", "LE"))

levels(correlaciones_estructura_pearson$Structure)

o <- ggplot(correlaciones_estructura_pearson, aes(x = tipo_celular,
                                                   y = Structure, fill = cor_pearson)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(cor_pearson, 2),
                color = cor_pearson < 0.5)) +
  scale_color_manual(values = c("TRUE" = "black", "FALSE" = "white"), guide = "none") +
  scale_fill_gradient2(low = "white", mid = "#a6c8ff", high = "#08306b", midpoint = 0.5,
```

```r
                       limits = c(0, 1), oob = scales::squish, na.value = "grey90") +
  scale_y_discrete(limits = rev(levels(correlaciones_estructura_pearson$Structure))) +
  labs(title = "Correlación de Pearson entre CIBERSORTx y BayesPrism",
       subtitle = "Células neoplásicas",
       x = "Tipo celular",
       y = "Estructura",
       fill = "Pearson r") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5))

print(o)
```

5. Fusionar células neoplásicas

```r
# Definir cuáles son las células neoplásicas
tipos_tumor <- c("AClike", "OPClike", "MES1like", "MES2like", "NPC1like", "NPC2like")

# ------ CIBERSORTX -------

ciber_grouped <- cibersort_lf_w_structure %>%
  mutate(
    muestra = as.character(muestra),
    Structure = as.factor(Structure),
    grupo_celular = ifelse(tipo_celular %in% tipos_tumor, "Tumor", tipo_celular)
  ) %>%
  group_by(muestra, Structure, grupo_celular, SubtypeW) %>%
  summarise(proporcion_cibersort = sum(proporcion_cibersort), .groups = "drop")

# ------- BAYESPRISM --------
bayes_grouped <- bayesprism_lf_w_structure %>%
  mutate(
    muestra = as.character(muestra),
    Structure = as.factor(Structure),
    grupo_celular = ifelse(tipo_celular %in% tipos_tumor, "Tumor", tipo_celular)
  ) %>%
  group_by(muestra, Structure, grupo_celular, SubtypeW) %>%
  summarise(proporcion_bayesprism = sum(proporcion_bayesprism), .groups = "drop")

# Unir los datos por muestra, estructura, grupo celular y subtipo
comparacion_grupo <- inner_join(ciber_grouped, bayes_grouped,
                                by = c("muestra", "Structure",
                                       "grupo_celular", "SubtypeW"))
```

5. Gráfico de microambiente tumoral

```r
# Calcular correlaciones de Pearson por estructura y grupo celular
correlaciones_grupo <- comparacion_grupo %>%
  group_by(Structure, grupo_celular) %>%
  summarise(cor_pearson = cor(proporcion_cibersort, proporcion_bayesprism,
                              method = "pearson"),
```

```r
    .groups = "drop")

# ------- Gráfico de dispersión --------
p <- ggplot(comparacion_grupo,
       aes(x = proporcion_cibersort, y = proporcion_bayesprism, color = Structure)) +
  geom_point(alpha = 0.6, size = 1.5) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +
  facet_grid(Structure ~ grupo_celular, scales = "free") +
  geom_text(data = correlaciones_grupo,
            aes(x = Inf, y = -Inf, label = paste0("r = ", round(cor_pearson, 2))),
            hjust = 1.1, vjust = -0.5,
            inherit.aes = FALSE,
            size = 3, color = "blue") +
  labs(x = "Proporción CIBERSORTx",
       y = "Proporción BayesPrism",
       title = "Correlación de Pearson entre BayesPrism y CIBERSORTx",
       subtitle = "Microambiente tumoral") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5),
        legend.position = "bottom")

ggsave(
  filename = "microambiente2.png",
  plot = p,
  path = "C:/Users/usuario/Documents/1. Practicas/resultados",
  width = 13, height = 6, dpi = 300,
  bg = "white")

print(p)
```

```r
# ----- Heatmap -------
correlaciones_grupo$Structure <- factor(
  as.character(correlaciones_grupo$Structure),
  levels = c("CT", "CTmvp", "CTpan", "IT", "LE"))

ggplot(correlaciones_grupo, aes(x = grupo_celular, y = Structure, fill = cor_pearson)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(cor_pearson, 2)), size = 3,
            color = ifelse(correlaciones_grupo$cor_pearson > 0.6, "white", "black")) +
  scale_fill_gradient2(low = "white", mid = "#a6c8ff", high = "#08306b", midpoint = 0.5,
                       limits = c(0, 1), oob = scales::squish, na.value = "grey90") +
  scale_y_discrete(limits = rev(levels(correlaciones_grupo$Structure))) +
  labs(title = "Correlación de Pearson entre CIBERSORTx y BayesPrism",
       subtitle = "Microambiente tumoral",
       x = "Grupo celular",
       y = "Estructura",
       fill = "Pearson r") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5))
```

6. Comparación por zona

```r
#Asegurar que la columna 'metodo' existe
if (!"metodo" %in% names(bayesprism_lf_w_structure)) {
  bayesprism_lf_w_structure$metodo <- "BayesPrism"}
if (!"metodo" %in% names(cibersort_lf_w_structure)) {
  cibersort_lf_w_structure$metodo <- "CIBERSORTx"}

#Renombrar columnas si es necesario
if ("proporcion_cibersort" %in% names(cibersort_lf_w_structure)) {
  cibersort_lf_w_structure <- cibersort_lf_w_structure %>%
    dplyr::rename(proporcion_comparativa = proporcion_cibersort)}
if ("proporcion_bayesprism" %in% names(bayesprism_lf_w_structure)) {
  bayesprism_lf_w_structure <- bayesprism_lf_w_structure %>%
    dplyr::rename(proporcion_comparativa = proporcion_bayesprism)}

#Combinar dataframes
datos_combinados <- bind_rows(bayesprism_lf_w_structure, cibersort_lf_w_structure)

#Crear una variable combinada del método (Bayesprism o Cibersort) y el subtipo
datos_combinados <- datos_combinados %>%
  mutate(metodo_subtype = paste(metodo, SubtypeW, sep = "_"))

#Asignación de colores
colores_puntos <- c(
  "BayesPrism_Classical" = "#1f77b4",
  "BayesPrism_Mesenchymal" = "#2ca02c",
  "BayesPrism_Proneural" = "#ff7f0e",
  "CIBERSORTx_Classical" = "#1f77b4",
  "CIBERSORTx_Mesenchymal" = "#2ca02c",
  "CIBERSORTx_Proneural" = "#ff7f0e")

#Bucle
for (structure in unique(datos_combinados$Structure)) {

  datos_filtrados <- datos_combinados %>%
    filter(Structure == structure)

  #Wilcoxon test
  test_resultados <- datos_filtrados %>%
    group_by(tipo_celular) %>%
    wilcox_test(proporcion_comparativa ~ metodo, p.adjust.method = "BH") %>%
    add_significance() %>%
    add_xy_position(x = "tipo_celular", dodge = 0.8)

  #Gráficos
  grafico <- ggplot(datos_filtrados, aes(x = tipo_celular,
                                         y = proporcion_comparativa)) +
    geom_boxplot(aes(fill = metodo), position = position_dodge(width = 0.8),
                 outlier.shape = NA, alpha = 0.5) +
    geom_jitter(aes(color = metodo_subtype, group = metodo),
                position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.8),
                alpha = 0.5, size = 1) +
    stat_pvalue_manual(test_resultados, hide.ns = TRUE,
```

```
                       tip.length = 0.01, label = "p.signif") +
    labs(title = paste(structure),
         y = "Proporción celular",
         x = "Tipo celular",
         fill = "Método",
         color = "Subtipo tumoral") +
    theme_light() +
    scale_fill_manual(values = c("BayesPrism" = "#1f77b4",
                                 "CIBERSORTx" = "#ff7f0e")) +
    scale_color_manual(
      values = colores_puntos,
      breaks = c("BayesPrism_Classical", "BayesPrism_Mesenchymal",
                 "BayesPrism_Proneural"),
      labels = c("Classical", "Mesenchymal", "Proneural")) +
    guides(fill = guide_legend(order = 1),
           color = guide_legend(order = 2)) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

  print(grafico)

}
```

7. Agrupar las células neoplásicas como "Tumor" para ver su proporción total

```
#Asegurar columna 'metodo'
if (!"metodo" %in% names(bayes_grouped)) {
  bayes_grouped$metodo <- "BayesPrism"
}
if (!"metodo" %in% names(ciber_grouped)) {
  ciber_grouped$metodo <- "CIBERSORTx"
}

#Renombrar columnas si es necesario
if ("proporcion_bayesprism" %in% names(bayes_grouped)) {
  names(bayes_grouped)[
    names(bayes_grouped) == "proporcion_bayesprism"] <- "proporcion_comparativa"
}

if ("proporcion_cibersort" %in% names(ciber_grouped)) {
  names(ciber_grouped)[
    names(ciber_grouped) == "proporcion_cibersort"] <- "proporcion_comparativa"
}
#Combinar data frames
datos_combinados <- bind_rows(bayes_grouped, ciber_grouped)

#Crear variable que combina método y subtype para colorear puntos
datos_combinados <- datos_combinados %>%
  mutate(metodo_subtype = paste(metodo, SubtypeW, sep = "_"))

#Asignar colores
colores_puntos <- c("BayesPrism_Classical" = "#1f77b4",
                    "BayesPrism_Mesenchymal" = "#2ca02c",
                    "BayesPrism_Proneural" = "#ff7f0e",
```

```r
                    "CIBERSORTx_Classical" = "#1f77b4",
                    "CIBERSORTx_Mesenchymal" = "#2ca02c",
                    "CIBERSORTx_Proneural" = "#ff7f0e")

for (cell in unique(datos_combinados$grupo_celular)) {

  datos_filtrados <- datos_combinados %>%
    filter(grupo_celular == cell)

  #Wilcoxon test
  test_resultados <- datos_filtrados %>%
    group_by(Structure) %>%
    wilcox_test(proporcion_comparativa ~ metodo, p.adjust.method = "BH") %>%
    add_significance() %>%
    add_xy_position(x = "Structure", dodge = 0.8)

  #Gráfico
grafico <- ggplot(datos_filtrados, aes(x = Structure, y = proporcion_comparativa)) +
  geom_boxplot(aes(fill = metodo), position = position_dodge(width = 0.8),
               outlier.shape = NA, alpha = 0.5) +
  geom_jitter(aes(color = metodo_subtype, group = metodo),
              position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.8),
              alpha = 0.7, size = 2) +
  stat_pvalue_manual(test_resultados, hide.ns = TRUE, tip.length = 0.01, label = "p.signif") +
  labs(title = paste(cell),
       y = "Proporción celular",
       x = "Estructura histológica",
       fill = "Método",
       color = "Subtipo tumoral") +
  theme_light() +
  scale_fill_manual(values = c("BayesPrism" = "#1f77b4", "CIBERSORTx" = "#ff7f0e")) +
  scale_color_manual(
    values = colores_puntos,
    breaks = c("BayesPrism_Classical", "BayesPrism_Mesenchymal",
               "BayesPrism_Proneural"),
    labels = c("Classical", "Mesenchymal", "Proneural")) +
  guides(fill = guide_legend(order = 1),
         color = guide_legend(order = 2)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
  print(grafico)
  grafico
}
```